TMDB Movie Analysis

What made 18 of the biggest box office hits succeed?



By Molo Munyansanga · November 2026

Key Takeaways

Five findings from the analysis

$2.94B

Highest revenue: Avengers: Endgame

12.3×

Avatar earned 12.3 times its budget, the best return

$1B+

Every movie made over $1 billion in profit

$7.9B

The Avengers: the most successful franchise

James Cameron earned the most of any director: $5.19B from Avatar and Titanic

About the Data

18 blockbusters from The Movie Database (TMDB)

18
movies analysed

16 + 2
franchise + standalone movies

1997–2019
release years

Movie details were downloaded from TMDB, a public online movie database. For each movie, the data includes its budget, box office revenue, audience ratings, popularity, genres, cast and crew.

Keep in mind

All 18 movies are blockbusters that each earned over $1.2B, so the results describe top hits, not movies in general. Ratings and popularity on TMDB change over time; these were collected in November 2026.

How the Analysis Was Done

Five steps, from raw data to findings



Collect

Downloaded the movie details from TMDB

Clean

Fixed formats, removed errors and checked for gaps

Explore

Looked at the overall shape of the data

Analyse

Ranked movies and compared franchises and directors

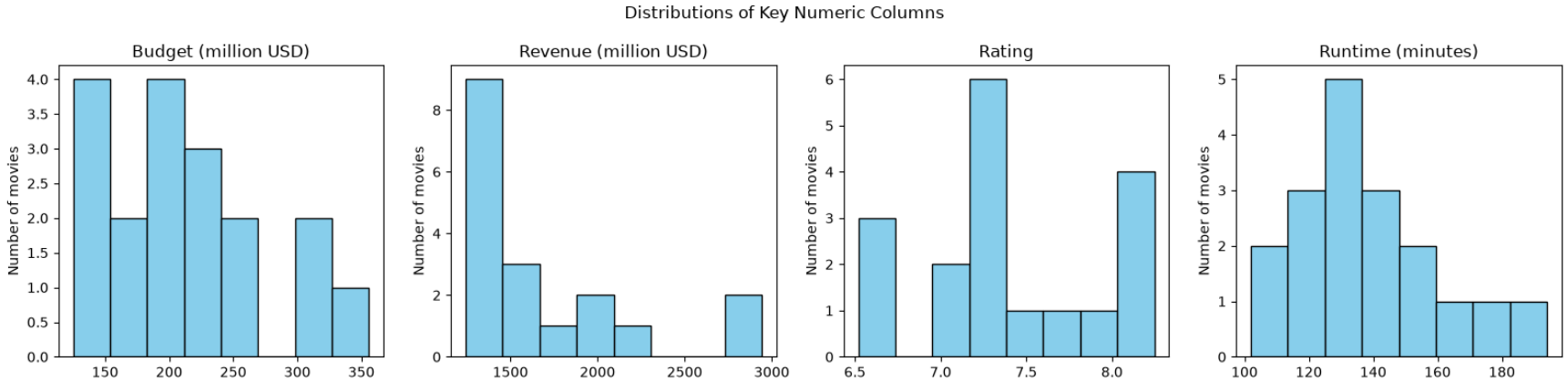
Visualise

Turned the findings into charts

Built with Apache Spark, a tool for processing very large datasets across many computers at once. The same code would work for millions of movies, and gave the same results as an earlier version built with Pandas.

What the Movies Look Like

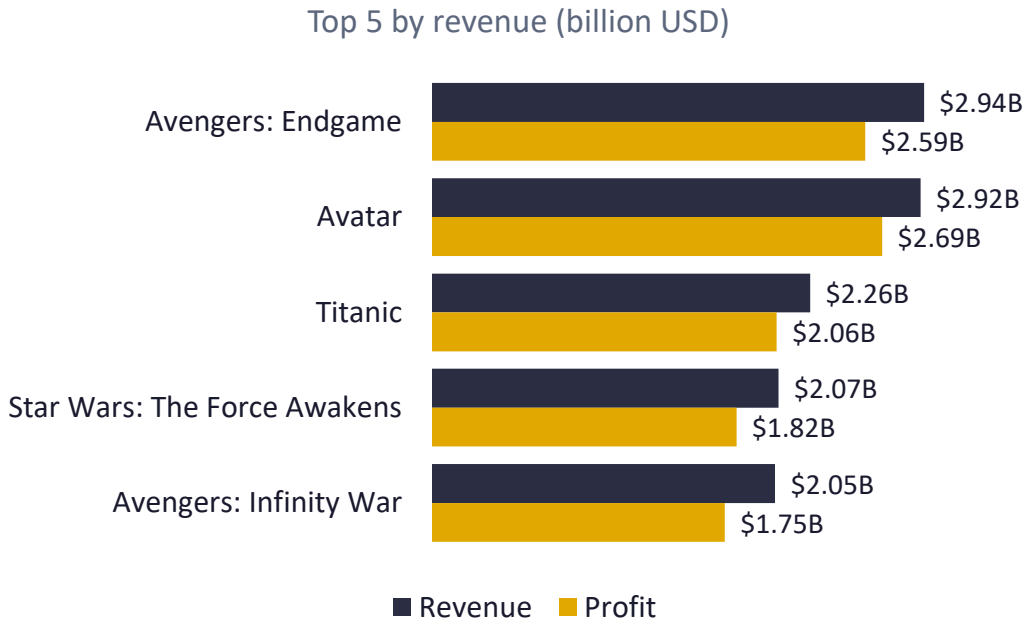
Most earned $1.2B–$1.45B; two earned close to $3B



- Half the movies (9) earned between $1.2B and $1.45B, while Avatar and Avengers: Endgame each earned about $2.9B.
- Most are rated between 7.2 and 7.4 out of 10, and run 120 to 140 minutes.

Top Earners

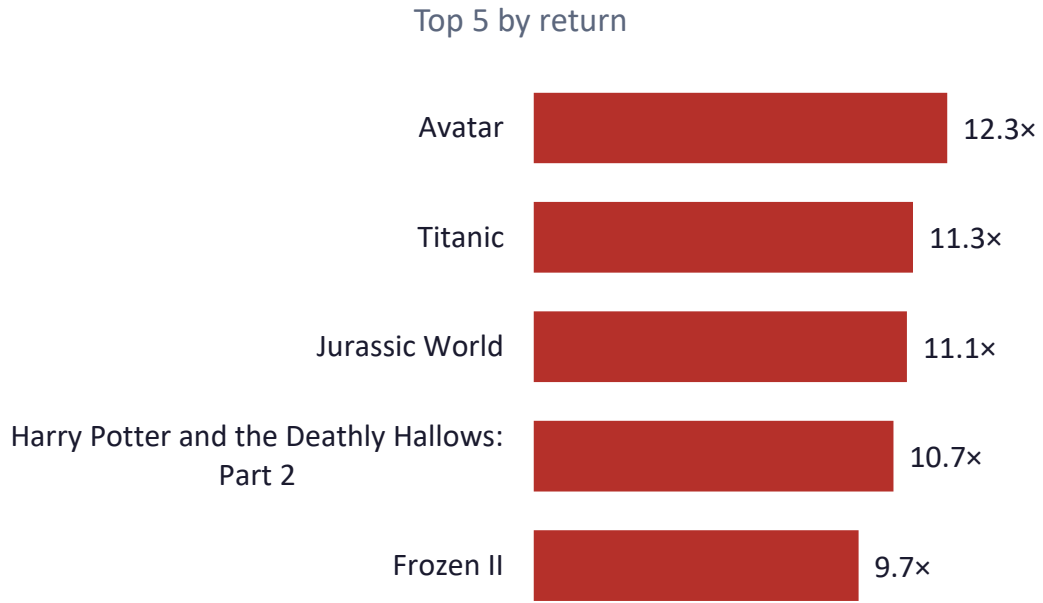
Endgame earned the most, but Avatar kept the most



- Avengers: Endgame had the highest revenue ($2.94B), just ahead of Avatar ($2.92B).
- Avatar cost far less to make ($237M vs. $356M), so it made the most profit: $2.69B.
- Titanic, the only standalone movie in the top 5, made the 3rd-highest profit ($2.06B).

Best Return on Investment

Return = how many times its budget a movie earned back



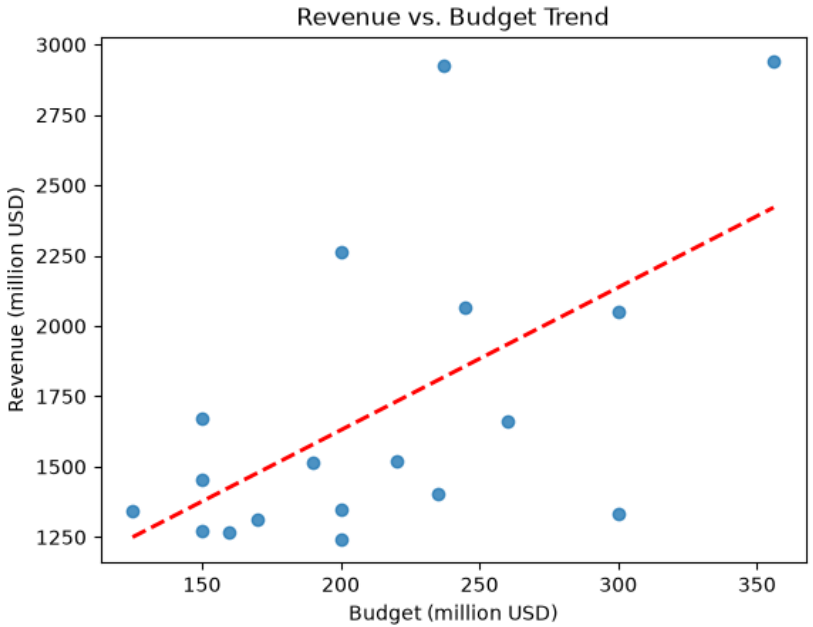
Lowest return

4.45×

Even Star Wars: The Last Jedi, the weakest performer, earned back its $300M budget more than 4 times.

Budget vs. Revenue

A bigger budget doesn't guarantee more revenue



- Revenue tends to rise with budget, but the movies are widely scattered around the trend line.
- Avatar earned $2.92B on a $237M budget, while The Last Jedi cost $300M and earned $1.33B.

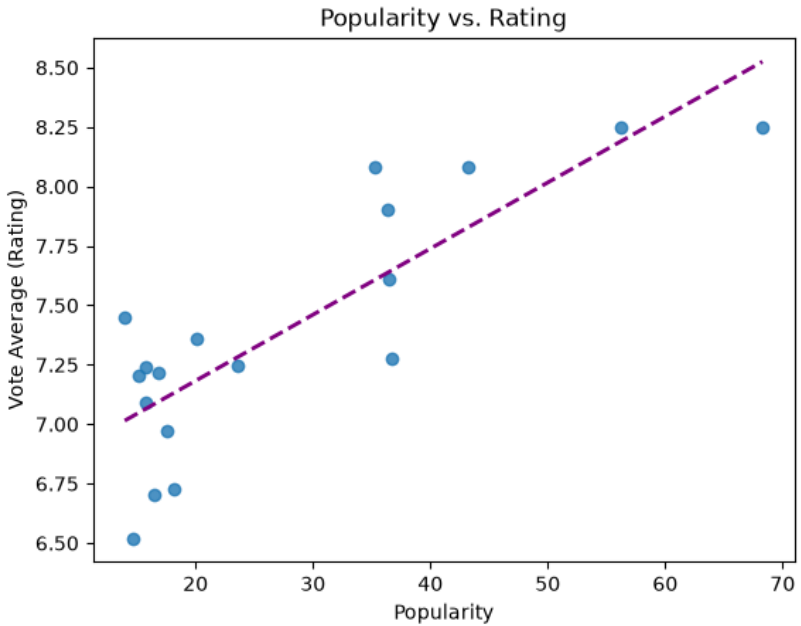
0.58

Link between budget and revenue (1 = perfect match): moderate

Popularity vs. Rating

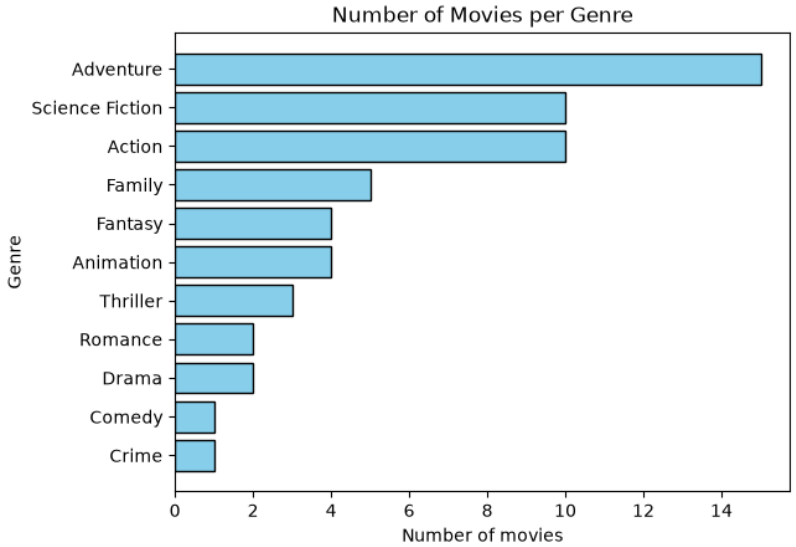
Audiences rate the most popular movies highest

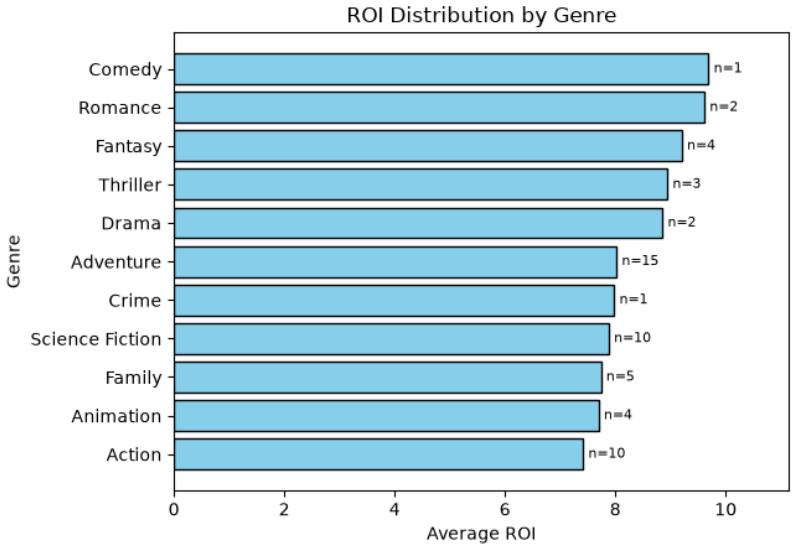
- The four most popular movies are all Avengers films.
- Avengers: Endgame and Infinity War are both the most popular and the highest rated (8.25 out of 10).
- This is the strongest link in the data: 0.83 on a scale where 1 is a perfect match.



Genres

Adventure is everywhere; returns are similar across genres



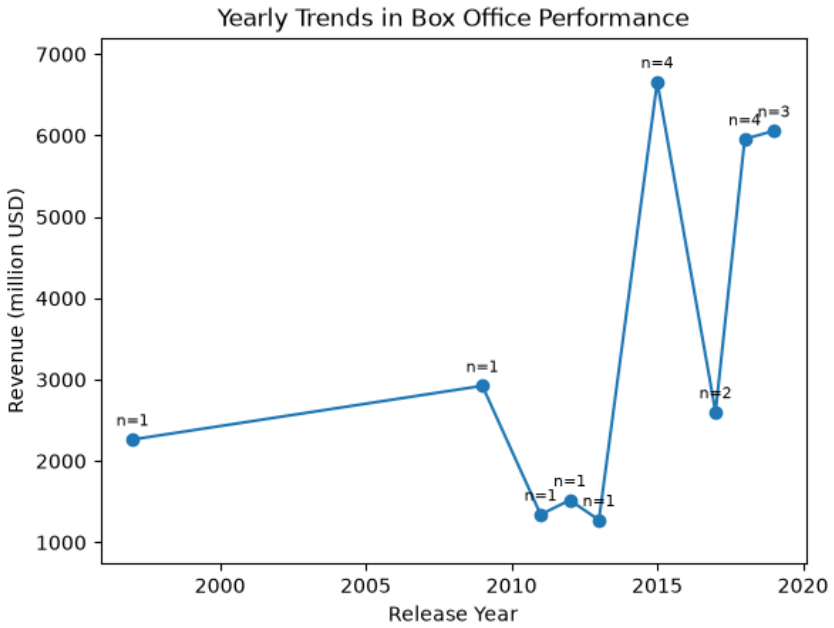


Adventure appears in 15 of the 18 movies; Action and Science Fiction in 10 each.

Comedy and Romance look best, but include only 1–2 movies (n), so they aren't reliable.

Box Office by Year

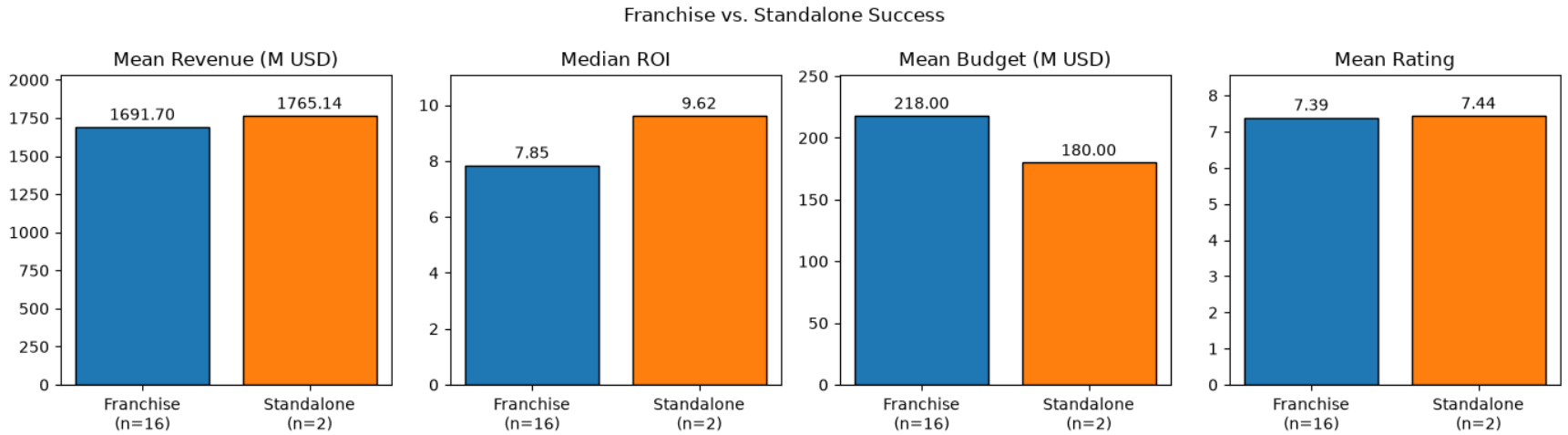
Peaks follow the number of releases, not a real trend



- Total revenue peaks in 2015 (about $6.7B) and 2018–2019 (about $6B).
- Those years simply had more movies in the dataset: 4 in 2015 and 2018, 3 in 2019 (n on the chart).
- There are no movies from 2014 or 2016 in the data.

Franchise vs. Standalone

No clear winner: only 2 movies are standalone



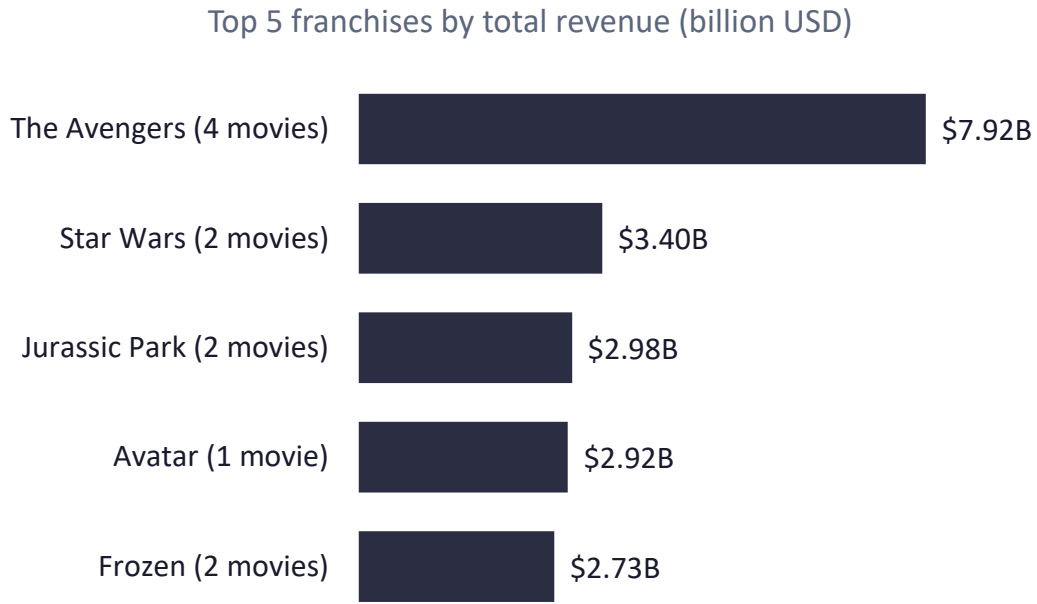
Franchise movies had bigger budgets: $218M vs. $180M on average.

Standalone movies earned slightly more revenue and return.

Only Titanic and Beauty and the Beast are standalone: too few to compare fairly.

Most Successful Franchises

The Avengers earned more than the next two franchises combined



$7.9B

from 4 Avengers movies, with an average rating of 7.96: the highest of any franchise with more than one movie.

Most Successful Directors

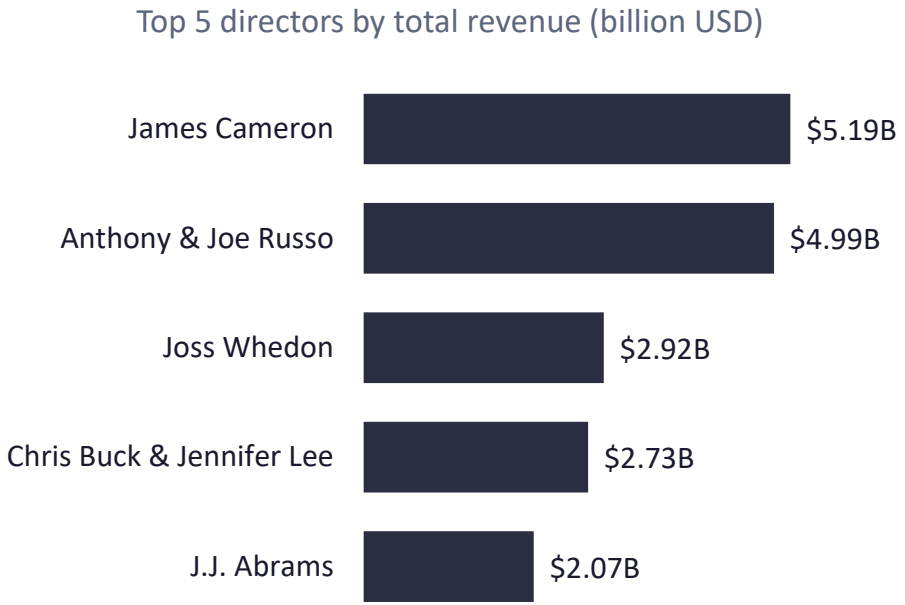
Most revenue vs. highest ratings

James Cameron

Most total revenue: $5.19B from Avatar and Titanic

Anthony & Joe Russo

Highest average rating (8.25) for Infinity War and Endgame



Conclusions

1

Collect more data

All 18 movies are blockbusters, so these results describe top hits, not movies in general.

2

Titanic stands out

3rd-highest profit ($2.06B) without a franchise or fantasy/sci-fi story. Worth studying how to repeat it.

3

Franchises: undecided

Franchises spent more and were slightly more popular, but 2 standalone movies are too few to compare.

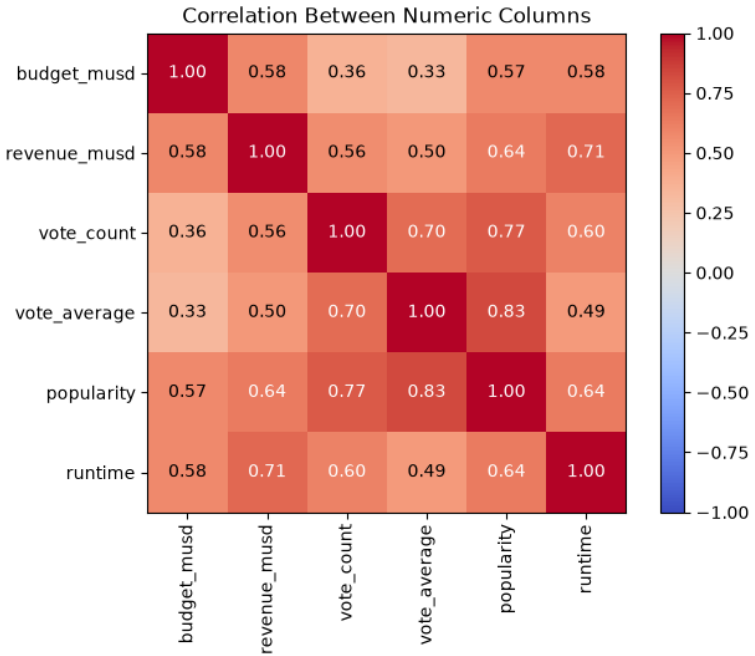
4

Money isn't everything

Budget is only moderately linked to revenue, while audiences rate the movies they love highest.

Appendix: Correlations

How strongly the numbers are linked: 1 = perfect link, 0 = no link



- Every measure moves in the same direction as the others (all links are positive).
- Strongest: popularity and rating (0.83). Popular movies are rated higher.
- Longer movies tended to earn more (0.71 between runtime and revenue).
- Budget has a moderate link with revenue (0.58) and a weak one with rating (0.33).